UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 002 · Solutions of homogeneous DE: complementary functions · compiled 15 October 2026

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1 Short notes

1 · The auxiliary equation and the complementary function
For $f(D)y = 0$ with constant coefficients, put $y = e^{mx}$: $f(D)e^{mx} = f(m)e^{mx} = 0$, and $e^{mx} \neq 0$, so $f(m) = 0$, the auxiliary (characteristic) equation. Write it by replacing D by m (or $y \to 1$, $y' \to m$, $y'' \to m^2$). Its roots decide the form of the complementary function, the general solution of the homogeneous equation, which has as many arbitrary constants as the order.

2 · Second-order equations: three cases
$y'' + k_1 y' + k_2 y = 0$, AE $m^2 + k_1 m + k_2 = 0$. Real distinct m_1, m_2: $c_1 e^{m_1 x} + c_2 e^{m_2 x}$ ($-3, 1$: $c_1 e^{-3x} + c_2 e^{x}$). Real equal m: $(c_1 + c_2 x)e^{mx}$ ($-2, -2$: $(c_1 + c_2 x)e^{-2x}$). Complex $\alpha \pm i\beta$: $e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$ ($2 \pm 3i$: $e^{2x}(c_1 \cos 3x + c_2 \sin 3x)$).

3 · Equations of order n: combining the cases
Order n: n roots, n constants. Each simple real root: ce^{mx}; a k-fold real root: $(c_1 + \cdots + c_k x^{k-1})e^{mx}$; each complex pair: $e^{\alpha x}(c \cos \beta x + c' \sin \beta x)$; a complex pair twice: $e^{\alpha x}[(c_1 + c_2 x)\cos \beta x + (c_3 + c_4 x)\sin \beta x]$; a zero root: a constant. Add the blocks. $y''' + 4y' = 0$: $c_1 + c_2 \cos 2x + c_3 \sin 2x$.

4 · Solving problems, with initial conditions
Initial values: write the general solution, differentiate, substitute, solve for the constants. CW: P3 $e^{2x}(2\cos x - 5\sin x)$; P4 $\frac{1}{2}(5e^{3x/2} - 3e^{x/2})$; P8 $e^x + 2e^{2x} - 3e^{3x}$; P9 $2\sin 2x + \sin 3x$. All nine CW answers check out.

2 Keyword index

20 terms; the number in brackets is the topic.

Arbitrary constants (1). As many as the order.

Auxiliary equation (1). $f(m) = 0$; also called the characteristic equation.

Auxiliary polynomial (1). $f(m)$.

Block (3). The CF part contributed by one root or pair.

Boundary conditions (4). Conditions at different points (CW P9 at 0 and $\frac{\pi}{2}$).

Complementary function (1). General solution of the homogeneous equation.

Complex (imaginary) roots (2). $e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$.

Conjugate pair (2). $\alpha \pm i\beta$; complex roots come in pairs.

Constant coefficients (1). $k_1, \ldots, k_n$ are numbers, not functions of x.

Discriminant (2). $k_1^2 - 4k_2$: sign picks the case.

Homogeneous LDE (1). $f(D)y = 0$, right-hand side zero.

Initial conditions (4). Values of $y, y', \ldots$ at one point: fix the constants.

k-fold root (3). Polynomial $c_1 + \cdots + c_k x^{k-1}$ times e^{mx}.

Multiplicity (3). How often a root repeats.

Particular solution (4). The general solution with the constants fixed.

Real and distinct roots (2). $c_1 e^{m_1 x} + c_2 e^{m_2 x}$.

Real and equal roots (2). $(c_1 + c_2 x)e^{mx}$.

Repeated complex pair (3). $(c_1 + c_2 x)\cos \beta x + (c_3 + c_4 x)\sin \beta x$.

Trial solution (1). $y = e^{mx}$.

Zero root (3). Gives a constant term.

3 Rapid recall

1	Trial solution for $f(D)y = 0$?	$y = e^{mx}$.
2	Auxiliary equation of $(D^2 + k_1 D + k_2)y = 0$?	$m^2 + k_1 m + k_2 = 0$.
3	Why can we cancel e^{mx}?	$e^{mx} \neq 0$ for every real x.
4	Another name for the AE?	Characteristic equation.
5	CF for real distinct roots m_1, m_2?	$c_1 e^{m_1 x} + c_2 e^{m_2 x}$.
6	CF for a double root m?	$(c_1 + c_2 x)e^{mx}$.
7	CF for $\alpha \pm i\beta$?	$e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$.
8	$y'' + 2y' - 3y = 0$?	$c_1 e^{-3x} + c_2 e^{x}$.
9	$y'' + 4y' + 4y = 0$?	$(c_1 + c_2 x)e^{-2x}$.
10	$y'' - 4y' + 13y = 0$?	$e^{2x}(c_1 \cos 3x + c_2 \sin 3x)$.
11	$y''' + 4y' = 0$?	$c_1 + c_2 \cos 2x + c_3 \sin 2x$.
12	Root repeated 3 times?	$(c_1 + c_2 x + c_3 x^2)e^{mx}$.
13	Complex pair repeated twice?	$e^{\alpha x}[(c_1 + c_2 x)\cos \beta x + (c_3 + c_4 x)\sin \beta x]$.
14	Number of arbitrary constants for order n?	n.
15	$m^4 + 4 = 0$ factorises as?	$(m^2 - 2m + 2)(m^2 + 2m + 2)$, roots $\pm 1 \pm i$.
16	$y^{iv} + 50y'' + 625y = 0$?	$(c_1 + c_2 x)\cos 5x + (c_3 + c_4 x)\sin 5x$.
17	CF of $(D^3 - 2D^2 + D)y = X$?	$c_1 + (c_2 + c_3 x)e^{x}$.
18	CW P8 answer?	$e^{x} + 2e^{2x} - 3e^{3x}$.
19	CW P9 answer?	$2\sin 2x + \sin 3x$.

Mnemonic (study aid). *"Distinct: add; equal: times x; complex: cos and sin."* The three cases in one line (study aid).

Mnemonic (study aid). *"k repeats, powers up to $k - 1$."* The polynomial in front of a repeated root (study aid).

4 Last 10 minutes before the exam

Top facts

1. $y = e^{mx}$ turns $f(D)y = 0$ into the AE $f(m) = 0$: replace D by m.
2. Distinct: $c_1e^{m_1x} + c_2e^{m_2x}$. Equal: $(c_1 + c_2x)e^{mx}$. Complex $\alpha \pm i\beta$: $e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$.
3. k-fold root: $(c_1 + \cdots + c_kx^{k-1})e^{mx}$. Repeated complex pair: $(c_1 + c_2x)\cos \beta x + (c_3 + c_4x)\sin \beta x$, times $e^{\alpha x}$.
4. Zero root gives a constant; $\pm i\beta$ gives $\cos \beta x$, $\sin \beta x$ with no exponential.
5. Order n = number of roots = number of constants.
6. Initial conditions: general solution → differentiate → substitute → solve for c's → check.
7. ESA: no homogeneous-only question, but all 24 Unit 2 question occurrences need a CF (table C4.2).

Top formulas

E1.2 · Auxiliary equation ESA ★★★

$$f(D)y = 0 \implies f(m) = m^n + k_1m^{n-1} + \cdots + k_n = 0$$

GIVES	The n roots $m_1, \ldots, m_n$ that fix the CF.
APPLIES	Homogeneous, constant coefficients.
NOTE	Replace D by m (equivalently $y \to 1$, $y' \to m$, $y'' \to m^2$, $y''' \to m^3$).

S002·5 · S002·10 · T002 00:03:18 · T002 00:07:40

E2.1 · Case I: real and distinct roots ESA ★★★

$$m_1 \neq m_2 \text{ real}: \quad y = c_1e^{m_1x} + c_2e^{m_2x}$$

GIVES	The CF.
APPLIES	Discriminant $k_1^2 - 4k_2 > 0$.
NOTE	Slide 002·7: $m = -3, 1$ gives $y = c_1e^{-3x} + c_2e^{x}$.

S002·7 · T002 00:05:40

E2.2 · Case II: real and equal roots ESA ★★★

$$m_1 = m_2 = m: \quad y = (c_1 + c_2x)e^{mx}$$

GIVES	The CF.
APPLIES	$k_1^2 - 4k_2 = 0$, $m = -\frac{k_1}{2}$.
NOTE	"This is the formula you should remember" (T002 00:09:10).

S002·8 · T002 00:08:57 · T002 00:09:10

E2.3 · Case III: complex roots ESA ★★★

$$m = \alpha \pm i\beta : \quad y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$$

GIVES The CF, in real form.

APPLIES $k_1^2 - 4k_2 < 0$; α = real part, β = imaginary part ($\beta > 0$).

NOTE $\alpha = 0$ (roots $\pm i\beta$) gives $c_1 \cos \beta x + c_2 \sin \beta x$.

S002·9 · T002 00:10:56

E3.2 · Case (ii): a root repeated k times ESA ★★★

$$y = (c_1 + c_2 x + c_3 x^2 + \cdots + c_k x^{k-1}) e^{mx}$$

GIVES The block of the CF for a k-fold real root m.

APPLIES Slide 002·12 has $k = n$ (all roots equal) and $k = 4$.

NOTE Stop at x^{k-1}: a double root gives $c_1 + c_2 x$, a triple root $c_1 + c_2 x + c_3 x^2$.

S002·12 · T002 00:14:58 · T002 00:15:15

E3.5 · Case (v): a complex pair occurring twice ESA ★★

$$y = e^{\alpha x}\left[(c_1 + c_2 x)\cos \beta x + (c_3 + c_4 x)\sin \beta x\right] + c_5 e^{m_5 x} + \ldots$$

GIVES The CF.

APPLIES $\alpha \pm i\beta$ repeated twice.

NOTE Both the cosine and the sine get a polynomial of degree 1 (CW P7).

S002·13 · T002 00:18:21 · T002 00:18:51

E3.7 · Zero root ESA ★★

$$m = 0 : \quad c\, e^{0x} = c$$

GIVES A constant term in the CF.

APPLIES $f(m)$ has a factor m (no y term in the DE).

NOTE Slide 002·14: $m = 0, \pm 2i$ gives $c_1 + c_2 \cos 2x + c_3 \sin 2x$. A double zero root gives $c_1 + c_2 x$.

S002·14 · T002 00:21:20